

IMPLEMENTATION OF A TRIAGE ALGORITHM TO ASSIGN ACUITY TO IMPROVE
ACCESS TO CARE AND IMPROVE WAITING TIMES

by

Michael Lindsey Brown

A Scholarly Project
Submitted to the Graduate School of
Northwestern State University of Louisiana
In partial fulfillment of requirements for the
Doctor of Nursing Practice degree

August 2023

IMPLEMENTATION OF A TRIAGE ALGORITHM TO ASSIGN ACUITY TO
IMPROVE ACCESS TO CARE AND IMPROVE WAITING TIMES

by

Michael Lindsey Brown

A DNP Project
Submitted to the Graduate School of
Northwestern State University of Louisiana
In partial fulfillment of requirements for the
Doctor of Nursing Practice Degree

Approved by:

Leah Pearce, DNP, RN

Leah Pearce, DNP, RN, CNE
Major Professor

Susan K. Steele-Moses, DNS, RN

Susan K. Steele-Moses, DNS, RN
CONSAH Statistician and Methodologist

Greg Handel/CH Digitally signed by Greg Handel/CH
Date: 2023.07.24 10:36:53 -05'00'

Greg Handel, DMA Date
Dean, The Graduate School

PREVIEW

Copyright 2023

Michael Lindsey Brown

ABSTRACT

IMPLEMENTATION OF A TRIAGE ALGORITHM TO ASSIGN ACUITY TO IMPROVE ACCESS TO CARE AND IMPROVE WAITING TIMES

Brown, Michael Lindsey, MSN, APRN, FNP
BSN, RN, Northeast Louisiana University, 1997
MSN, Grambling State University, 2012
Doctor of Nursing Practice, Northwestern State University, Fall Commencement 2023
Major: Organizational, Systems, Leadership: Doctor of Nursing Practice
*Implementation Of A Triage Algorithm To Assign Acuity To
Improve Access to Care And Improve Waiting Times*
Scholarly Project directed by Professor Dr. Leah Pearce
Pages, 127. Words in abstract, 303.

Purpose: The purpose of the DNP Quality Improvement project was to apply Lean Six Sigma principles to eliminate waste and duplication while establishing a framework for implementing the ESI triage algorithm. The main objective was to enhance the clinic's throughput, decrease inefficiencies, and improve access to care. The specific Occupational Medicine Clinic (OMC) involved in this project did not utilize a triage or acuity-based system, which led to bottlenecks, extended wait times, and chronic inefficiency.

Procedure: The project team followed Lean Six Sigma principles and utilized an evidence-based framework Define-Measure-Analyze-Improve-Control (DMAIC) to successfully implement the ESI triage algorithm from January 16th to March 10th. The following clinical question was developed to address the clinic's challenges: In an occupational medicine clinic, will implementing the ESI triage algorithm using Lean Six Sigma principles improve access to care and clinic wait times compared to the traditional practice in eight weeks?

Results: Overall analysis, using a two-way Analysis of variance (ANOVA), revealed significant differences when comparing the pre-and post-implementation groups rooming wait times between the pre-and post-implementation groups $F(1, 174) = 1284.971, p = .014$, partial $\eta^2 = .034$; however, no difference based on the ESI acuity level. Continued analysis revealed, when assigning acuity, statistically significant differences in the total visit times based on ESI scores and total visit times by Group. When considering the interaction between Group and ESI scores, a significant difference was also observed ($F(2, 174) = 10710.74, p < .001$, partial $\eta^2 = .137$), and in summary, the total visit times demonstrated significant differences based on ESI scores, Group, and their interaction.

Conclusions: The framework of DMAIC and the use of Lean Sigma principles provided the guidance needed to implement the ESI algorithm successfully. The use of the ESI algorithm positively impacted the time a patient was in the clinic, improving access to care.

Acknowledgments

I would like to take this opportunity to express my heartfelt gratitude to the following individuals who have been instrumental in supporting me throughout my DNP journey. First and foremost, I am deeply grateful to my loving wife, Tara Brown, for her unwavering support and understanding. Her patience, encouragement, and belief in my capabilities have been a constant source of motivation. Without her by my side, this accomplishment would not have been possible. I would also like to acknowledge my two sons, Davis Brown and Tyler Brown, whose love and understanding during this challenging period have been invaluable. Their belief in my pursuit of higher education has inspired me to persevere and reach my goals. I owe my Aunt, Stella Jean Brown, a debt of gratitude for her wise counsel and unwavering support. Her words of encouragement always came at the right time, helping me navigate the pressures of school, work, and life with renewed determination.

I extend my sincere appreciation to the management and staff of St. Francis Medical Center for their continuous support and understanding. Their willingness to accommodate my educational pursuits while working full-time has been a tremendous blessing. Their commitment to fostering a culture of lifelong learning has contributed significantly to my growth and development as a healthcare professional.

I also extend my most profound appreciation to my Major Professor. Over the past year, Dr. Leah Pearce's mentorship fostered growth that I did not realize was possible. Dr. Pearce's understanding and continued encouragement when I recently lost my mother provided comfort and said I would persevere through this difficult time.

I am forever indebted to my parents, Milton L. Brown and Holly Brown, for instilling in me the values of perseverance, hard work, and compassion. My father's teachings to never give

up and to keep striving until I reach my goals have been a guiding force in my life. My mother's lessons on compassion and staying grounded have shaped me into who I am today. Though Aunt Stella and my Mother are no longer with us, I know that they are watching over me as guardian angels, cheering me on as I take these final steps in my DNP journey.

Finally, I would like to express my sincere appreciation to all my family members, including my sidekicks Louis and Roux. All of you have supported and believed in me throughout this endeavor. Your love, encouragement, and belief in my abilities have been a constant source of strength. I hope that I have made you proud with this accomplishment. Thank you all for being integral to my journey and enriching my life in countless ways.

TABLE OF CONTENTS

Copyright 2023	iii
ABSTRACT	iv
Acknowledgments	vi
List of Tables	xi
List of Figures	xii
CHAPTER ONE: INTRODUCTION	1
Background of the Problem	1
Statement of the Problem	3
Clinical Question	4
Participants of Interest	4
Practice Change	5
Comparison	5
Purpose, Aims, and Objectives	6
Significance of the Project	7
Needs Assessment	7
SWOT (Strengths, Weaknesses, Opportunities, and Threats) Analysis	12
Target Population and Stakeholders	14
Conceptual and Theoretical Framework	14
Define	15
Measure	15
Analyze	16
Improve	16
Control	16
Definition of Terms	17
Occupational Medicine Clinic	17
Bottlenecking	17
Wait Time	17
Triage	18
Access to Care	18
Acuity	18
Assumptions, Limitations, and Delimitations	19
Assumptions	19
Limitations	19
Delimitations	20
Summary	20
CHAPTER TWO: SYNTHESIS OF EVIDENCE	21
Search Strategies	21
Levels of Evidence	22
Synthesis of Evidence	24
Wait Times and Length of Stay (LOS)	24
Perceptions of Staff and Patients	28
Emergency Severity Index (ESI)	30
Lean Six Sigma Principles	32
Gaps in Knowledge	34

Summary	35
CHAPTER THREE: METHODOLOGY	36
Purpose of the Scholarly Project.....	36
Population and Sample Selection.....	37
Sampling Strategy and Site Authorization	37
Expenses and Source of Funds	38
Instrumentation and Project Data.....	39
Data Collection Procedures.....	40
Data Analysis Procedures	42
Ethical Considerations	42
Respect for Persons	43
Benevolence	43
Justice	44
The Scientific Review Committee (SRC) and Institutional Review Board (IRB) Approval Process.....	44
Summary	44
CHAPTER FOUR: RESULTS	46
Important Changes and Updates.....	47
Preparation of Raw Data for Analysis	48
Data Analysis Procedures	48
Results.....	49
Outcome One - Check-in Wait Times	49
<i>Descriptive Findings</i>	49
<i>Tests of Assumptions</i>	50
<i>Inferential Statistics</i>	54
Outcome Two – Rooming Wait Times	54
<i>Descriptive Findings</i>	54
<i>Tests of Assumptions</i>	56
<i>Inferential Statistics</i>	58
Outcome Three – Total Visit Times.....	59
<i>Descriptive Findings</i>	59
<i>Tests of Assumptions</i>	61
<i>Inferential Statistics</i>	63
Summary	64
CHAPTER FIVE: SUMMARY, CONCLUSIONS, AND RECOMMENDATIONS	65
Summary and Discussion of Results.....	66
Project Results Related to Clinical Question	67
Practice Implications.....	68
Theoretical and Conceptual Framework.....	69
Define	70
Measure	70
Analyze.....	70
Improve.....	70
Control.....	71
Strengths	71
Limitations.....	71

Recommendations	71
Future Implications	71
Sustainability	73
Dissemination Plans	74
Conclusions	74
Appendix A	86
Appendix B	87
Appendix C	88
Appendix D	89
Appendix E	90
Appendix F	91
Appendix H	101
Appendix I	102
Appendix J	103
Appendix K	104
Appendix L	105
Appendix M	106
Appendix N	107
Appendix O	108
Appendix P	109
Appendix Q	110
Appendix R	111
Appendix S	114
AFFIRMATION OF ACADEMIC HONESTY	115

List of Tables

Table 1	10
Table 2	13
Table 3	15
Table 4	23
Table 5	39
Table 6	48
Table 7	50
Table 8	51
Table 9	54
Table 10	55
Table 11	58
Table 12	58
Table 13	59
Table 14	60
Table 15	63
Table 16	63
Table 17	64

List of Figures

Figure 1	12
Figure 2	41
Figure 3	51
Figure 4	52
Figure 5	52
Figure 6	53
Figure 7	53
Figure 8	55
Figure 9	56
Figure 10	57
Figure 11	60
Figure 12	61
Figure 13	62
Figure 14	69

CHAPTER ONE: INTRODUCTION

According to Edwards (2013), triage has a long history dating back to the 18th century during the Napoleonic Wars. The chief battlefield surgeon coined the French word "trier" or triage, meaning "separate out." Triage is typically associated when addressing emergency room (ER) processes and is used to coordinate care for those with the most urgent needs. According to Tam et al. (2018), incorrectly using the triage system can lead to a patient safety event. Like ER processes, outpatient settings also struggle with many of the same barriers including long wait times. Long wait times create a chokepoint, sometimes called a bottleneck. A bottleneck in healthcare is defined as anything causing a restraint or restriction preventing patients from completing the clinic or healthcare visit timely. Outpatient settings, including walk-in clinics, urgent care centers (UCCs), and occupational medicine clinics (OMCs), are three examples of outpatient settings that suffer from the same barriers as the ER. Naiker et al. (2018) described that identifying inefficiencies will improve quality and access when delivering care. Implementing a triage system in an occupational outpatient clinic can assist in resource alignment, operational efficiency, and strengthen clinic processes.

Background of the Problem

No matter the setting, triage is essential to deliver timely care. Crowding in the waiting room is a problem that emergency rooms, urgent care clinics, and walk-in clinics contend with and continue to attempt to solve. Kang et al. (2014) found that implementing alternative admission processes could reduce patient length of stay (LOS) and positively affect patient flow. Occupational medicine clinics, ERs, and UCCs, struggle to find tools to improve efficiency and wait times. Rocha and Vasconcelos (2021) recommended following a standardized process to reduce total waiting time. Appropriate training is warranted to understand the process of triaging

patients to improve efficiency and decrease wait times. Occupational medicine providers administer services for employers and companies throughout the United States. However, in today's healthcare environment, a health system must deliver high-quality care from the moment a customer walks over the clinic's threshold until the customer is given an after-visit summary on discharge (Vashi et al., 2018).

Lot et al. (2018) explained that extended wait times would negatively affect patient outcomes, perceptions, satisfaction scores, employee burnout, and quality of care. DeAnda (2018) discussed how ERs continue to increase in visits requiring patients to remain in the ER until the patients are discharged due to patient rooms or beds not being available or because the staff is not available to provide care.

Similarly, in 1980 UCCs were steadily being developed, with thousands of facilities being built nationwide (Yee et al., 2013). UCCs provide care on a walk-in basis and can be closely aligned with OMCs. UCCs provide care for common complaints such as on-the-job injuries, animal bites, auto accidents, dermatology issues caused by employment environment, burns, vision-threatening conditions, minor to significant lacerations, and fractures (Rothstein et al., 2021). Chokepoints or bottlenecks in healthcare can occur in departments primarily serving walk-in patients, usually based on a lack of throughput processes (Arya et al., 2013). When there are limited processes to manage patient surges, the backup of patients into the waiting room can become significant.

Improving a clinic or emergency room's process has been researched using several quality improvement methodologies. Emergency rooms and clinics are not the only healthcare industry departments plagued with long wait times. Finding a specific benchmark for occupational medicine clinics related to wait times can be difficult because occupational clinics

are not mandated to report measures or outcomes to a governing body. Ambulatory Health Care Facilities and Occupational Health Clinics can pursue accreditation voluntarily; however, The Joint Commission may impose reporting requirements in the next few years (Taylor, 2022). Occupational medicine visit types are closely related to the urgent care population and can be compared with other clinics associated with waiting times and efficiency.

Bottlenecks are not a new problem and have been studied for years, attempting to improve patient throughput using the appropriate triage system. Assigning appropriate acuity classifications and implementing quality improvement initiatives can improve patient flow in UCCs (DeAnda, 2018). Poor processes can cause lengthy wait times and affect patient safety, patient satisfaction, employee satisfaction, and business loyalty (Medical Group Management Association [MGMA], 2022). Efficiency is an essential concept for nursing, no matter what the topic of interest, even with quality improvement and processes being poorly executed. According to Khomami and Rustomfram (2019), nursing is responsible for effective communication with patients and other healthcare providers. Nurses are labeled as the mainstay of health care and must comply with standards of care when ethical concerns arise. When efficient, cost-effective, and timely care is provided, the length of stay decreases, and patient satisfaction is enhanced.

Statement of the Problem

The problem statement provides a succinct description of the problem; however, it should explain why it needs to be evaluated and, if the problem has merit, why it needs to be corrected (Moran et al., 2019). Correcting a situation can only occur once a problem has been identified. "Waiting" is a widespread problem across healthcare industries, specifically the ER, UCC, and walk-in clinics. Waiting times have been a problem for years and are a continuous struggle for healthcare. According to Burström et al. (2013), the average wait time in an ER setting is greater

than four hours and even longer for older patients seeking care. Patients waiting for extended periods will lead to poor satisfaction and increased frustration. Prolonged wait times in healthcare can occur when a process fails to move patients successfully throughout the healthcare cycle (DeAnda, 2018). A thorough investigation of clinic efficiency will provide the knowledge needed to change the burdensome wait time dilemma and, in turn, improve throughput by using a standardized system of managing care. Kovach and Ingle (2019) explained that there were two ways to measure quality in outpatient clinics, including a patient's wait time and the number of no-shows, understanding that no-shows usually occur due to expecting to wait.

Clinical Question

The clinical question is derived from the problem statement, which helps the project manager describe the problem and clarify any ambiguities (Moran et al., 2019). Using a PICOT format is effective in developing a concise clinical question. A PICOT question includes the following attributes: Population, Intervention, Comparison, and Outcomes over a specific period (Moran et al., 2019). The PICOT question for this DNP quality improvement project was: In an occupational medicine clinic, will implementing the ESI triage algorithm using Lean Six Sigma principles improve access to care and clinic wait times compared to the traditional practice in eight weeks?

Participants of Interest

The OMC patients are candidates for employment, injured employees, or employees who require testing for compliance with Occupational Safety and Health Administration (OSHA). The target population for this DNP QI project was patients who frequented the specified OMC located in Northeastern Louisiana during the project period from sign-in to discharge.

Approximately 40-50 potential job candidates in Northeast Louisiana visited the OMC daily. The OMC has provided care for employers in the Southeast Arkansas region, Northeast Louisiana region, and third-party administrators across the nation authorizing potential and current employees to have testing or treatment at the OMC.

According to LANP (2022), 95% of all residents in Louisiana live in a primary care health professional shortage area. Medicaid covers 40% of residents in Louisiana, 72.2% of Louisiana Nurse Practitioners deliver primary care, 80% care for Medicaid patients, and less than 12% of Louisiana physicians are primary care providers. The vulnerable population for the DNP QI project was the patients that visited the OMC. Most of this population were uninsured or on Medicaid, relying on ERs and UCCs to provide most of their care. Based on these statistics, access to care in Louisiana remains an issue.

Practice Change

This DNP quality improvement project aimed to determine if implementing the ESI triage algorithm effectively improved wait times and access to care by assigning acuity for all patients that arrived for treatment, testing, or follow-up. Since most of the OMC population have Medicaid or are uninsured, the OMC continues to be an entry point to a provider-patient relationship. It encourages these individuals to obtain care through primary care instead of ERs or UCCs. The opportunity to provide care, even during a pre-employment exam, allows the provider to determine if referrals or additional services are needed. The OMC can continue to navigate this vulnerable population to appropriate providers to improve health care outcomes.

Comparison

In the identified OMC, there was no triage process to determine patient acuity and throughput order. Prior to implementation, the patients were prioritized by the time of arrival

regardless of visit type or patient needs. The lack of a standardized process related to triage and coordinating care caused increased waiting room times, and patient safety can be at risk. OMC's process included a mixture of electronic and paper charting, leaving the staff of the OMC to find a strategy to improve efficiency and patient flow.

The registration process was lengthy, taking approximately ten to 15 minutes to complete each registration. Bottlenecking occurred when multiple patients walked in simultaneously without an appointment. Employees in the OMC were expected to anticipate others' needs and potential problems while communicating with the provider and co-workers. Employees who did not fully understand their role could create barriers and cause other performance issues (Harris et al., 2020).

Purpose, Aims, and Objectives

The purpose of the DNP quality improvement project was to use quality improvement principles, such as Lean Six Sigma, to eliminate waste and duplication and provide a framework for ESI triage implementation. This DNP project leader aimed to improve the clinic's throughput and decrease waste and duplication by implementing the ESI triage algorithm. This specific OMC participated in the DNP quality improvement project and did not have a triage or acuity-based system. The current documented clinic process caused bottlenecking, extended wait times, and chronic inefficiency. Following Lean Six Sigma principles and an evidence-based framework, the team successfully implemented the triage algorithm to assign an acuity level to each patient and improve access to care in an occupational medicine clinic within eight weeks. According to Moran et al. (2019), a DNP should ground their project using a theoretical framework for guidance and understanding.

The DNP quality improvement project objectives were to decrease wait times and to

improve access to care by assigning acuity to the walk-in population to improve clinic flow and throughput. With a successful implementation of the ESI triage tool and assignment of appropriate acuity, the goals were to decrease clinic waiting times, including sign-in to check-in times, check-in to provider, and total clinic time until discharge.

Significance of the Project

Healthcare has been transforming continuously since the COVID-19 pandemic. Healthcare systems, urgent care clinics, and walk-in clinics have been redesigned to improve the delivery of care and healthcare services. This OMC in Northeast Louisiana followed the same path. With competitors providing occupational medicine services in the area, OMC must improve processes and the quality of work to be recognized as the first choice for employers in the occupational medicine market. When patient throughput is improved, patient wait times will decrease, and more patients can be seen, increasing revenue, decreasing non-productive time, and improving access to care.

Needs Assessment

According to Moran et al. (2019), conducting a needs assessment at the "macro-level" will allow the project manager to see a broader picture and identify potential problems for the area assessed. The project manager can take the "macro-level" assessment's findings and perform a more precise "micro-level" assessment to ascertain the population's immediate needs. Team members should take an active part in the Quality Improvement (QI) process from the inception allowing for employee empowerment and advancing new knowledge while improving operations. Vashi et al. (2018) explained that one needs to include efficiency, effectiveness, and timeliness to obtain high-quality care. Performing a needs assessment at the clinic's site was critical to determine if a need exists and if an improvement was warranted.

For a QI project, a gap includes three categories based on an individual clinic's needs. These categories include knowledge, skills, or practice. According to Golden et al. (2017), using an effective tool will succinctly recognize inconsistencies in care delivery. Gaps in health care delivery can be addressed using Lean tools, such as value stream mapping, to validate the current flow process in the clinic. Lean principles encourage team members to participate (George et al., 2005).

A needs assessment or gap analysis followed the Lean Six Sigma principles for this DNP Quality Improvement project. Lean Six Sigma or Lean principles provided a suitable pathway for healthcare to recognize and redesign failing processes (Hung et al., 2019). The needs assessment started with the planning process to give a baseline of the current clinic status. Preparing for the project was essential before initiating the Define-Measure-Analyze-Improve-Control (DMAIC) process. Lean Six Sigma principles guided the QI project, but completing a needs assessment was a priority. Preparation had several facets, including how the project leader managed the proposed practice change, understanding the possibility of resistance to the proposed change, and identifying the requirements for the staff. The project leader needed to recognize that change can be stressful and painful but also understand that it was a necessity for healthcare to be sustainable (Sperl et al., 2013).

According to Agency for Healthcare Research and Quality (2015), the employee and stakeholder questionnaire provided clinicians and nonclinical staff tools to improve the quality of care and patient safety for office-based clinics, including specialty offices. According to Sperl et al. (2013), using the *Lean Healthcare Assessment and Gap Analysis* questionnaire provided the staff an opportunity to give valuable feedback to leadership, creating a baseline and an understanding of possible necessary process changes. Surveys or questionnaires are effective

ways to obtain needed data and should be succinct (Harris et al., 2020).

The Lean Six Sigma needs assessment questionnaire contained 29 questions categorized into five sections measuring leadership, patient focus, process management, staff management, and information analysis (Sperl et al., 2013). Choosing the correct tool was essential to decide on the clinic's needs. The project leader performed a needs assessment at the clinic's site. This assessment was critical to determine if a need existed and if an improvement was warranted. The 29-question survey was slightly adapted from a standard survey used in other Lean Six Sigma projects to document the perceptions of the team members and stakeholders of the OCM (see Appendix A).

The survey was entered into Survey Monkey, and the survey link was emailed to all OCM staff. The open-ended question, added to the standard survey, allowed the participants to specify the one change they would recommend. Two questions were added to the beginning to identify their years of service to OMC and job titles. The OMC employees and stakeholders who were invited to take the survey included one clinic manager, two patient access representatives (PARs), four clinical staff, two nurse practitioners, an employee health supervisor, a registration supervisor, two senior directors, the current executive leadership of the clinic, and the outpatient physical therapy manager.

Patient access representatives were the first to address the patients in the clinic and were responsible for setting the tone and process for the patient's journey through the OMC. The team members' role requires a straightforward process and excellent communication skills. Clinicians, including Nurse Practitioners (NPs), Registered Nurses (RNs), and License Practice Nurses (LPNs), had an essential role in the back office once a patient was called back to be seen. The needs assessment results were provided using a Lean Sigma Radar Graph tool (see Appendix B).

The stakeholders' comments can be found in Appendix C. The needs assessment ranked the answers using a Likert scale, determining the lowest-scoring question in each section. Each area provided Lean Six Sigma recommendations that assisted with improvement to each area (see Table 1).

Table 1

Gap Analysis Recommendations for Improvement

Assessment Topics	Question	Recommendations
Leadership: Resources	• Adequate resources are continuously assigned, achieving service excellence.	• Develop a standardized method • Discuss with staff realistic goals. • Work with staff and perform continuous improvement.
Customer Focus: Benchmarking	• The organization obtains and uses customer service information relative to benchmarks.	• Begin to develop plans for improved customer service measures. • Develop a plan to benchmark customer service.
Process Management: Efficiency and Variation	• The processes that were used are smooth and work well. • Process variation is understood and monitored at the organization.	• Implement performance measures to track the efficiency of clinic processes. • Provide improved communication and performance measures. Allow staff to participate in problem-solving. • Consider training and education sessions on process variation. Implement statistical process control techniques in critical areas of the organization. • Conduct regular process variation review sessions.
People Management: Development	• Leadership creates an encouraging and motivating environment and helps people reach their full potential.	• Define the current clinic environment and discuss potential improvements. • Develop an improvement plan and begin implementation. • Develop a communication plan and receive constructive feedback from employees.
Information and Analysis: Visibility	• The monitoring system is clear and visible to the organization (e.g., scoreboards).	• Discuss how to make the newly implemented QI and performance measures for others to review • Implement standardized methods to oversee new improvements and QI measures

Note. The noted recommendations validate the need for a standardized system to improve clinic processes. Lean Six Sigma's needs assessment and gap analysis process provide a pathway for quality improvement (Sperl et al., 2013).

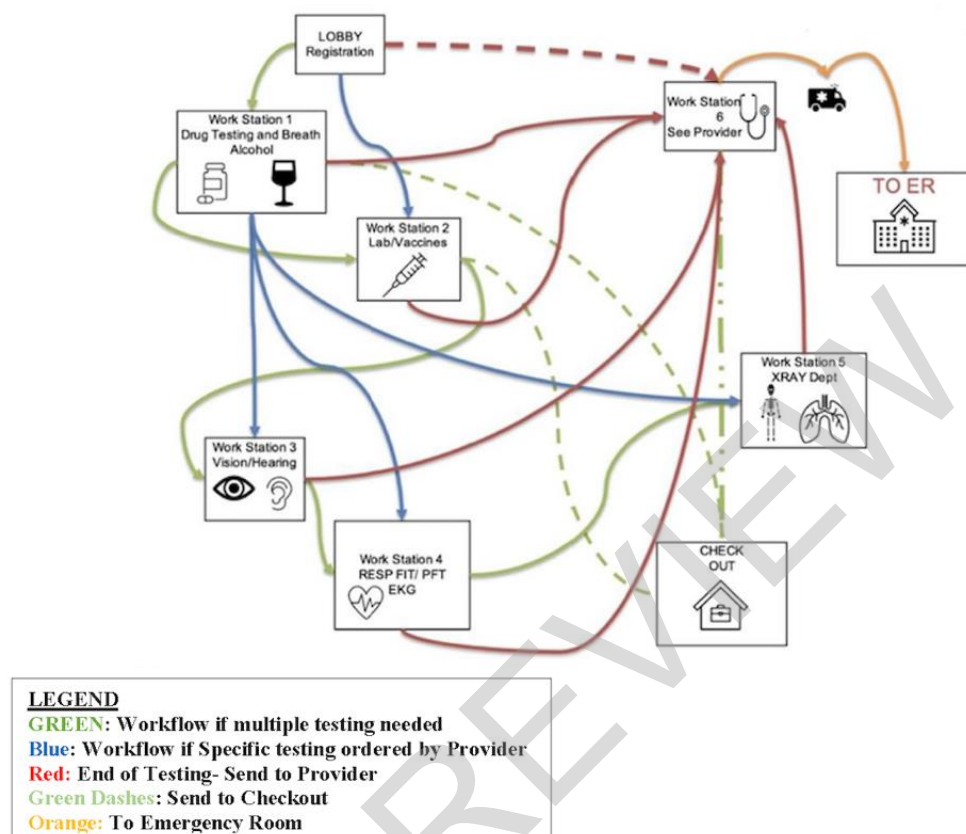
When the needs assessment questionnaire was distributed and completed, the project manager used an additional Lean method to identify waste and duplication. This method was called a Gemba Walk. Lot et al. (2018) stated that managers in the healthcare setting have had to redesign and develop a new pathway, such as a process map or flowchart, using lean methods like the Gemba Walk. Gemba, one of Lean Six's premiere tools, originated in Japan. Gemba means to walk through the "workplace or where the work occurs" (Simon & Canacari, 2011, p. 86). The Gemba Walk allowed the observer to be immersed in current processes and document the patient flow from the beginning to the end of the patient encounter. A Gemba Walk was used to follow the patient through the clinic, collect information, and document clinic flow and clinic processes (Sampath et al., 2021).

After completing the Gemba Walk in the clinic, the level of disorganization was apparent. The clinic received paper faxes instead of a shared folder on Outlook. The disorganization was caused due to lack of standardization in the workflow and an antiquated communication process. Two significant workflow issues were noted, the loss of faxed documents and the loss of authorization to test documents; both caused inefficiency and delays in the registration process. The observer noted that the PARs who completed the registration and check-in process acknowledged that shortcuts were being used. The registration and paperwork were placed in a rack with no identifiable acuity listed. The clinic staff would call the patient to the back, based on the order of registration completion, without consideration of acuity. The Gemba Walk process allowed the team to collect needed information regarding the process flow and potential causes of inefficiencies and delays. A spaghetti diagram was used to depict the clinic flow, signifying congestion and potential problems with lines crisscrossing multiple times (see Figure 1). The completed Gemba Report and template can be found in Appendix D, which

was used to record the clinic's workflow (Continuous Improvement Toolkit [CIToolkit], 2022).

Figure 1

Spaghetti Diagram



Note. Spaghetti diagram workflow adapted from Lean Six Sigma (George et al., 2005).

The Gemba Walk determined a chaotic flow to the clinic and cannot be blamed on one specific employee or area of the clinic. All team members participated, which caused the chaotic flow. It was noted that improved communication processes and teamwork were needed to correct the broken system. According to George et al. (2005), the project-team approach allows team members to work on the project while continuing regular job duties.

SWOT (Strengths, Weaknesses, Opportunities, and Threats) Analysis

A SWOT analysis was used with a needs assessment to compare the current practice to the potential practice change (Harris et al., 2020). OCM team members and stakeholders